

Project Report: Week 1

Ethical Data Sourcing, Preprocessing, and Exploration

1. Introduction

The objective of Week 1 was to collect, inspect, clean, and preprocess an anonymized heart disease dataset for healthcare predictive analysis. All analysis was performed on publicly available and de-identified medical data to ensure ethical handling of healthcare information and maintain patient privacy.

2. Dataset Collection

The dataset was obtained from an open-source medical dataset repository and contains clinical and cardiovascular health-related attributes such as age, cholesterol level, blood pressure, ECG results, heart rate, and heart disease condition.

The dataset includes the following important medical biomarkers:

- Age
- Gender
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol
- Fasting Blood Sugar
- Resting ECG
- Maximum Heart Rate
- Exercise Induced Angina
- ST Depression
- ST Slope
- Major Vessels
- Thalassemia
- Heart Disease

3. Data Inspection

Initial inspection of the dataset was performed using:

- `df.head()`
- `df.info()`
- `df.describe()`

These methods helped identify:

- dataset structure
- column names
- datatypes
- missing values
- summary statistics

4. Missing Value Handling

Missing values were identified using:

```
df.isnull().sum()
```

Imputation Strategy

Numerical medical features were handled using **median imputation**, while categorical variables were handled using **mode imputation**.

Clinical Rationale

Median imputation was selected for numerical medical biomarkers because it is less sensitive to outliers and extreme physiological values commonly observed in healthcare datasets. Mode imputation was applied to categorical variables to preserve the most frequent clinical category and maintain dataset consistency.

5. Duplicate Record Handling

Duplicate patient records were checked using:

```
df.duplicated().sum()
```

Any duplicate records identified were removed to improve data quality and ensure reliable analysis.

6. Data Preprocessing

The dataset was preprocessed by:

- renaming column headers for readability
- handling missing values
- validating datatypes
- removing duplicate records
- preparing the dataset for exploratory analysis and machine learning

7. Summary Statistics

Descriptive statistical analysis was generated using:

```
df.describe()
```

This provided insights into:

- mean
- median
- minimum and maximum values
- standard deviation
- distribution of medical biomarkers

The preprocessing phase improved dataset quality by handling missing values, correcting data types, and removing duplicate records. These preprocessing steps ensured that the dataset became clean, consistent, and suitable for future medical data analysis and machine learning workflows.